MILITARY SPECIFICATION

FILTER, GAS, 5000 CFM, C23

1. SCOPE

1.1 This specification covers a gas filter for use in removing toxic gases from the air forced or drawn through the filter.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-P-530 - Plywood, Flat Panel.
TT-E-527 - Enamel, Synthetic, Lustreless.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

CHEMICAL CORPS

C5-19-2052 - Filter, Gas, 5000 CFM, C23 - Bill of Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply:

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing C5-19-2052 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the item shall conform to the specifications and drawings listed on Drawing C5-19-2052 and subsidiary drawings.

3.2 Manufacture and assembly. The filter shall be manufactured, assembled and marked as shown on Drawing D5-19-2015 and subsidiary drawings, which are listed on Drawing C5-19-2052.

3.3 Leakage. The filter shall not leak when tested as specified in 4.5.1.

3.4 Air resistance. The air resistance of the painted filter shall not exceed 1.50 inches of water when tested as specified in 4.5.2.

3.5 Workmanship. The filter shall be free from contamination (foreign matter), defects or damage that might impair its use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all materials and components are in accordance with 3.1 and section 5 (see 6.3).

4.3 Alternate inspection (including testing) procedures. The supplier may utilize any alternate inspection procedures which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 Lotting. A lot shall consist of the filters produced by one manufacturer from the same materials and under essentially the same manufacturing conditions. However, not more than two lots of C21 250 cubic feet per minute (c.f.m.), gas filters shall be represented in any one lot of C23 filters.

4.4.2 Sampling. Sampling for nondestructive inspection shall be conducted in accordance with Standard MIL-STD-105. The level and degree of inspection shall be specified in the contract.

4.4.3 Nondestructive examination. The regular production lot shall be inspected in accordance with the classification of defects and Standard MIL-STD-105.

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4.4.4 Classification of defects.

4.4.4.1 Filter, gas, 5000 c.f.m., C23-assembly. (Dwg. D5-19-2015)

Categories and defects	Acceptance standards
Critical	
*A. Leakage	4.5.1
*B. Air resistance	4.5.2
Major — AQL 1.0 percent defective	
1. Components missing, incorrect or incorrectly assembled	
2. Damage affecting functioning	
3. Adhesive material incorrectly applied	
4. Marking incorrect	
Minor — AQL 2.5 percent defective	
51. Contamination	
52. Marking missing or illegible	

*Each filter in the lot shall be tested for leakage and air resistance.

4.4.4.2 Preparation for delivery. (section 5).

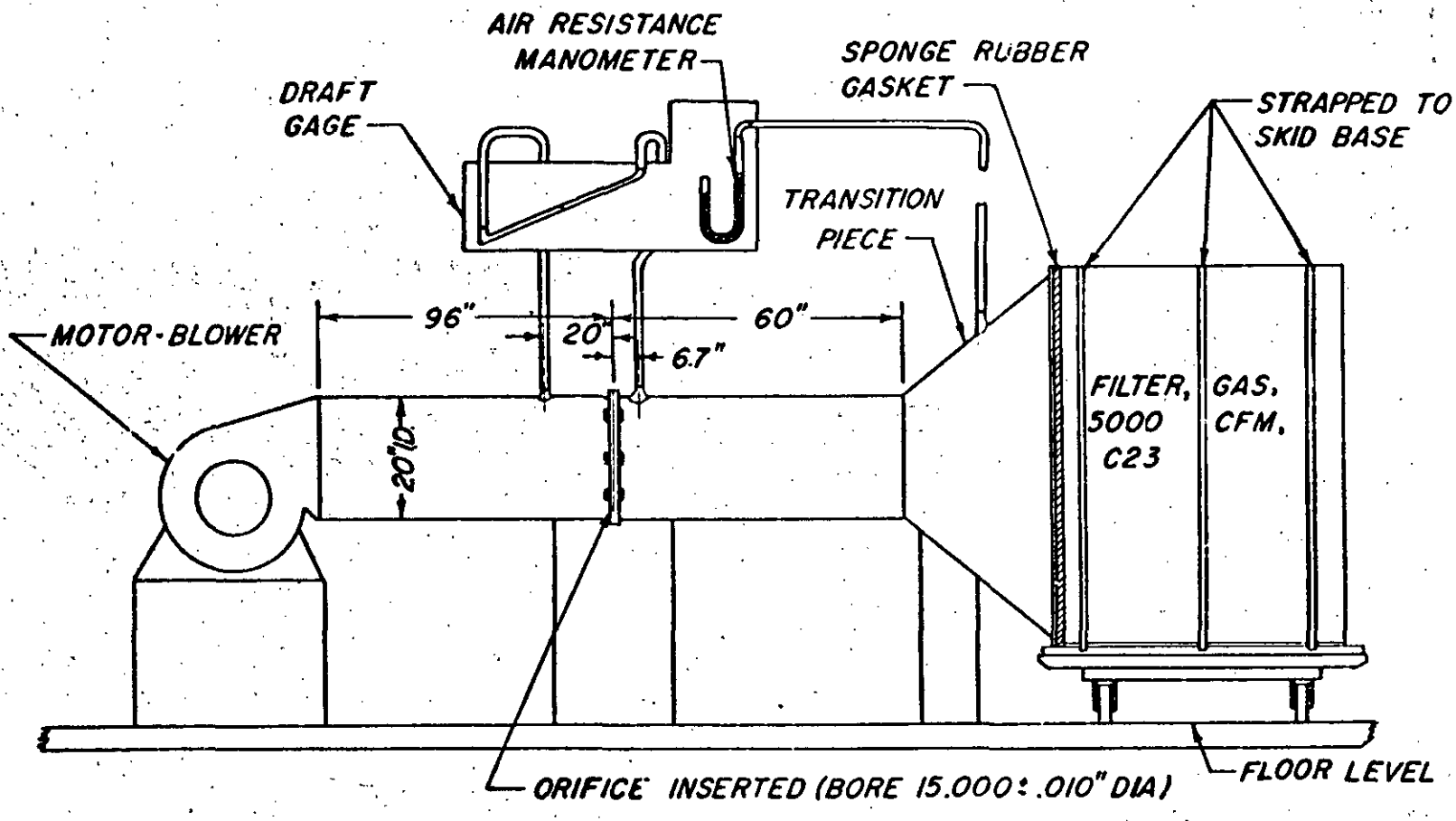
Categories and defects	
Major — AQL 4.0 percent defective	
1. Inlet and outlet openings of the filter not covered as specified	
2. Crate type, class or style incorrect	
3. Anchoring, blocking or bracing of filter to crate base missing	
4. Marking incorrect, missing or illegible	

4.5 Tests.

4.5.1 Leakage. The filter shall be tested for leakage as follows:

(a) Connect the filter to a source of forced air as shown on figure 1.
(b) Regulate the airflow to the influent ducts of the filter to 5000 c.f.m. by means of the orifice and draft gage.
(c) Spray isoamyl acetate (using a spray gun) into the inlet of the motor blower at a uniform rate of 1 ounce per minute for 5 minutes.

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SCALE: NONE

NOTE:

1. SECURE FILTER TO FLANGE OF TRANSITION PIECE WITH 4 LONG CARPENTER'S CLAMPS, PRIOR TO TURNING ON MOTOR-BLOWER.
2. SILVER BRAZE ALL TAPS FLUSH WITH INSIDE OF DUCT.

Figure 1 - Test apparatus.

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(d) Wear a gas mask facepiece equipped with a hose and hold the open end of the hose against each effluent duct, in turn, and test the air by sniffing while moving the hose along the duct. Regulate the movement of the hose so that a thorough sampling of the effluent air will be obtained in the 5-minute period specified in (c) above. Detection of the odor of isoamyl acetate indicates leakage.

(e) If leakage is detected, repair leak and retest using above procedure. No more than three leakage retests of one filter shall be permitted.

4.5.2 Air resistance. With the filter connected as shown on figure 1, the resistance shall be read on the manometer at an airflow of 5000 c.f.m.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. The inlet and outlet openings of the filter shall be covered with fitted 48 inch square by 1/4 inch thick exterior type, grade A-B, flat panel Douglas fir plywood, conforming to Specification NN-P-530, nailed in place. The plywood coverings shall be coated with synthetic lustreless enamel, color green, number 34087, conforming to Specification TT-E-527. The filter shall then be packed in a type II, class 1, style a crate conforming to Specification MIL-C-104, with the filter securely anchored, blocked and braced to the base of the crate in accordance with the appendix thereto. Strapping used shall be flat galvanized strapping.

5.1.2 Level B. Not applicable.

5.1.3 Level C. With the inlet and outlet openings of the filter protected as specified in 5.1.1, the filter shall be packed for shipment in accordance with the manufacturer's commercial practice in such manner as to provide protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers used shall comply with the Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The filter covered by this specification is a component of Filter, Gas and Particulate, 5000 CFM, M17 which is used for protecting fixed installations against toxic gases and aerosols.

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6.2 Ordering data. Procurement documents should specify:

(a) The title, number and date of this specification.
(b) The level of packing required.

6.3 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.4 Pilot lot. A pilot lot may be required to evaluate the supplier's production and quality control systems. The contracting officer will specify whether a pilot lot is required and applicable pilot lot acceptance criteria.

6.5 Supersession data. This specification supersedes Chemical Corps purchase Description 197-54-579A dated 29 October 1958.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

Army - Chemical Corps

Preparing activity:

Army - Chemical Corps

GPO 829619

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